

HOMEWORK ASSIGNMENT 6 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

QUESTION 1: Strands

Laura has recently gotten into the *New York Times* game **Strands**, in which you're presented with a grid of letters and are tasked with finding some number of words all linked by a common theme. (The exact specifics of the game aren't too important for this problem.) In *Strands*, if you get stuck, you can request a "hint." Suppose we model each of Laura's games as a trial of a random process, and suppose she's interested in constructing a confidence interval for the long-run proportion of times she needs to use a hint, a quantity we can call π_{Hint} . Among the 100 games she last played, she used a hint in 29 of them.

- (a) Describe one observational unit, in the context of this study.

Solution: One observational unit is one game of Strands.

- (b) Navigate to the [One Proportion Applet](#), and use it to fill in the following table. Use a simulation-based approach to calculate the p -values, and use at least 10,000 repetitions each time.

Proportion under H_0 :	0.20	0.21	0.22	...	0.38	0.39
p -value	0.0325	0.0632		...	0.0633	0.0407
Reject or Plausible?	Reject	Plausible	Plausible	...	Plausible	Reject

Table 1: Resulting p -values, and whether each specified value is plausible for π or not.

Solution: See table above; exact p -values may vary slightly.

- (c) Report the 95% confidence interval for π_{Hint} , obtained using the Plausible Values Method.

Solution: (0.21, 0.38).

- (d) Calculate and report a 95% confidence interval for π_{Hint} using the 2SD method. **Hint:** you may need to go back to the One Proportion Applet.

Solution: (0.21, 0.38).

- (e) Are the validity conditions met? Explain.

Solution: Yes. We have 29 successes, which is greater than or equal to 10; furthermore, we have 71 failures, which is also greater than or equal to 10.

- (f) Construct a theory-based 95% confidence interval for π_{Hint} **without** using the Theory-Based Inference Applet.

Solution:

$$\begin{aligned}\hat{p} \pm 2 \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} &= 0.29 \pm 2 \cdot \sqrt{\frac{0.29 \cdot (1 - 0.29)}{100}} \\ &= 0.29 \pm 2 \cdot 0.045 \\ &= 0.29 \pm 0.09 = \mathbf{(0.20, 0.38)}\end{aligned}$$

- (g) Navigate to the [Theory-Based Inference Applet](#), construct a theory-based 95% confidence interval for π_{Hint} , and check that it is close to your answer to part (f) above.

Solution: CI: (0.2011, 0.3789)

- (h) Interpret the confidence interval you obtained in part (g) above, in the context of the study.

Solution: We are 95% confident that the true long-run proportion of times Laura needs to use a hint in a game of Strands is between 20.11% and 37.89%.

- (i) **Without doing any calculations, or using any applets**, how would the width of a 90% confidence interval for π_{Hint} compare to the width of the interval you constructed in part (f) above? Explain.

Solution: We know that decreasing the confidence level also decreases the width of the corresponding confidence interval. As such, a 90% confidence interval for π_{Hint} should be **narrower** than a 95% confidence interval for π_{Hint} .

QUESTION 2: Gaga for Gaga

What is the average duration (in minutes) of a Lady Gaga song? To answer this question, Luka has taken a simple random sample of 40 Lady Gaga songs and recorded the durations (in minutes) of each of these songs. Shown below is a histogram of Luka's data, along with some summary statistics.

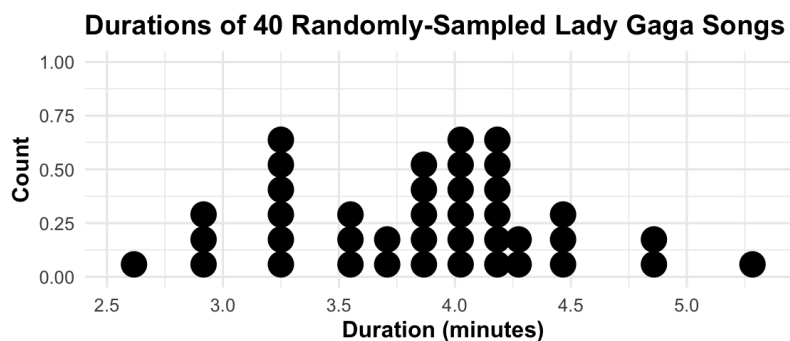


Figure 1: Results from Luka's data.

Sample Mean	Sample Median	Sample Standard Deviation	Sample Size
$\bar{x} = 3.86$ mins.	median = 3.94 mins.	$s = 0.58$ mins.	$n = 40$

Table 2: Summary statistics of the data from Luka's Data.

- (a) What does one dot on Figure (1) represent, in the context of this study?

Solution: One dot is one song and its duration in minutes.

- (b) Define the parameter in words and in symbols, using the context of this study.

Solution: The parameter of interest is the average duration, in minutes, of all Lady Gaga songs, which we denote by μ .

- (c) Are the validity conditions met? Explain.

Solution: Because the variable is quantitative, we use the validity conditions for a one-sample t -test. First we note that the sample size (40) is greater than equal to 20; then we note that the sample distribution (shown in Figure (1) above) is not too heavily skewed. Therefore, the validity conditions **are** met.

- (d) Construct a theory-based 80% confidence interval for the parameter of interest. Do not use an applet; instead, you may use the fact that the appropriate multiplier is around 1.304.

Solution:

$$\begin{aligned}\bar{x} \pm 1.304 \cdot \frac{s}{\sqrt{n}} &= (3.86) \pm 1.304 \cdot \frac{0.58}{\sqrt{40}} \\ &= (3.86) \pm 1.304 \cdot (0.0917) = (3.86) \pm (0.12) = \mathbf{(3.74, 3.98)}\end{aligned}$$

- (e) Interpret the confidence interval you obtained in part (d) above, in the context of the study.

Solution: We are 80% confident that the true average duration, in minutes, of a Lady Gaga song lies between 3.74 minutes and 3.98 minutes.

- (f) It turns out that the true average duration, in minutes, across all Lady Gaga songs (excluding remixes) is 3.724 minutes. Is this value contained in the interval you constructed in part (d) above? Is this a problem? Explain.

Solution: **No**, the value of 3.724 is not included in our interval. However, this is **not** a problem, because we know that confidence intervals are not guarantees! That is, due to **sampling variability**, we happened to get a sample for which the true value of the parameter was not contained in the 80% confidence interval. Indeed, if we repeat this procedure for constructing an 80% confidence interval using different random samples of Lady Gaga songs we'd expect the true value of the parameter (3.724 minutes) to be included in around 80% of the intervals.

QUESTION 3:

The following parts are unrelated.

- (a) Studies conducted in New York City and Boston have noticed that more heart attacks occur in December and January than in all other months. Some people have tried to conclude that holiday stress and overindulgence causes the increased risk of heart attack. Identify a confounding variable whose effect on heart attack rate might be confounded with that of the month.

- ☐ Stress due to work
- ✓ **Snow shoveling**
- ☐ Living in a large city
- ☐ Climate

Solution: Remember that a confounding variable should be associated with both the explanatory variable and the response variable. In particular, a confounding variable is one whose potential effects on a response variable cannot be distinguished from those of the explanatory variable. **Note:** an argument could technically be made for “Stress due to work” being a confounding variable whose effect on heart attack rate might be confounded with that of the month - on an open-ended question, I’d mainly look for correct logic/justification.

- (b) An analysis of data on a sample of 22 countries shows a strong positive association between the average life expectancy in the country and the number of TVs per 1,000 people in the country, that is, countries that have more TVs per 1,000 people also have higher average lifetimes. Based on this study, is it reasonable to conclude that by sending TVs to countries with lower life expectancies, we can increase their inhabitants’ lifetimes?
- ☐ Yes ✓ **No**
- (c) A randomized experiment allows for the possibility of drawing a cause-and-effect conclusion between _____ and _____
- ✓ **The explanatory variable and the response variable**
- ☐ The observational units and the variables
 - ☐ The subjects and the treatments
 - ☐ Statistical significance and statistical confidence
- (d) Which of the following must happen in a study to allow us to determine cause and effect?
- ☐ Simulation-based inference techniques
 - ☐ Taking a random sample from a population
 - ☐ Theory-based inference techniques
 - ✓ **Randomly assigning the observational units to different treatment groups**
- (e) **True or False:** Random sampling is a more important consideration than random assignment if the research question is whether faculty tend to drive older cars than students drive on your campus.
- ✓ **True** ☐ False
- (f) **True or False:** Random sampling is a more important consideration than random assignment if the research question is whether students tend to receive higher scores on essays if they are encouraged to submit a draft than if they are not so encouraged.
- ☐ True ✓ **False**